

Obstructions to Adelic Polarization

Part II of *Adelic Polarization Conjecture*

Matthew Long

The YonedaAI Collaboration, YonedaAI Research Collective

Chicago, IL

matthew@yonedaai.com

September 2026

Abstract

Arithmetic polarization through semilocal adelic spaces requires prime transitions that preserve arithmetic comparison as well as a geometric sign. We examine two particular ways of imposing these requirements. The ambient polar normalization of a known weighted entire-function carrier divides boundary values by the modulus of a local Euler factor. We prove that this operation cannot send any nonzero entire function to an entire function. The proof uses a simple zero of the squared Euler modulus and the parity of vanishing orders. The intrinsic polar decomposition on a closed Sonin subspace has a compressed metric operator and is not excluded by this argument. We derive its adjoint and explain the additional test-map and symmetry identities it must satisfy. A separate obstruction comes from exact compact smooth tests whose two pole moments vanish. At the arithmetic prime 11, three and four translated bumps give local quadratic values proportional to $-46/11$ and $(28\sqrt{11} - 92)/11$. These opposite signs rule out an orthogonal nonpositive enlargement with the prescribed positive local Weil increment on all intermediate tests. A finite packet calculation extends the sign result to every prime. The obstruction does not apply to sign imposed only at support-complete stages, or to correspondences with mixed pairings and explicit correction terms. The resulting formulation retains those terms and leaves the geometric sign and full arithmetic comparison as construction problems.

1 Introduction

A topological identification of semilocal Sonin spaces does not identify their inner products. This distinction matters when those spaces are used as a possible analytic realization of arithmetic cohomology. The construction of Connes, Consani and Moscovici provides a common space of entire functions with norms that change when a prime is adjoined [7, Sections 4.7 and 4.8]. An arithmetic polarization would need more than equivalent norms: its pairing must arise from arithmetic duality and reproduce the signed local terms of the explicit formula.

For a prime p , put $L = \log p$ and $r = p^{-1/2}$. In the common carrier, the finite change of weight is multiplication by

$$\omega_p(t) = |1 - re^{-itL}|^2 = 1 + r^2 - 2r \cos(tL). \quad (1)$$

This positive function lies between $(1 - r)^2$ and $(1 + r)^2$. The multiplier $\omega_p - 1$ is denoted d_p in Part I. The positivity of ω_p gives bounded comparison of the two norms. It does not give a sign to their difference, and it does not give a sign to the logarithmic derivative that produces the local Weil distribution. Part I establishes these local analytic identities with a fixed Fourier convention [9].

One possible repair is to replace the carrier identity by its polar isometry. In the ambient weighted L^2 spaces this means division by $\sqrt{\omega_p}$. The operation is well-defined, bounded and invertible on boundary functions. Theorem 3.4 shows that it destroys the entire-function condition for every nonzero input. Squaring a putative boundary identity produces an identity of entire functions. Its two sides have incompatible orders at zero. This obstruction does not depend on the growth or reproducing kernel of the Sonin space.

Polar decomposition within a closed Sonin space is a different operation. The adjoint of the restricted carrier identity includes the orthogonal projection onto that space. The associated positive operator is a compression of multiplication by ω_p , and its inverse square root need not be pointwise multiplication. We calculate this operator and its transformed projection in Proposition 4.1. The calculation preserves a possible intrinsic normalization while making its required arithmetic compatibility precise.

A second proposed repair is geometric: keep the old primitive sector isometrically and add an orthogonal nonpositive sector at each prime. The new sector then contributes a nonpositive quadratic increment. The required intersection increment, with the convention $B_W = -I$, is $+W_p$. That increment has both signs, even after the two pole moments vanish. We give exact smooth witnesses at $p = 11$, then a finite packet proof for every prime. The proof evaluates the actual distribution on autocorrelations; a finite matrix computation alone would not establish this statement about smooth tests.

The resulting exclusion has a specific support hypothesis. A new prime contributes zero to tests whose support was already complete for the old finite place set. The positive witness therefore obstructs the orthogonal sign ansatz on a nontrivial intermediate transition. It does not contradict a construction requiring its final sign only after all primes relevant to the support have been included.

Neither obstruction establishes a sign for the full Weil form. They reject two specified transitions and identify conditions a replacement must meet. The entire carrier, the local explicit formula and the cohomological motivation all precede this work. We make no claim that these elementary obstructions have priority over every treatment of Euler factors or compressed operators.

2 Mathematical framework

The conventions of Part I, Definition 2.1, are used throughout. Thus $\mathcal{T} = C_c^\infty(\mathbb{R}, \mathbb{C})$, convolution uses Lebesgue measure, $f^*(x) = \overline{f(-x)}$, and the centered Laplace transform is $F_f(z) = \int_{\mathbb{R}} f(x)e^{zx} dx$. The Fourier transform is $\hat{f}(t) = F_f(-it)$. These are imported conventions; the geometric test correspondence considered later is not part of their definition.

Part I, Definition 3.1, fixes $W = P - A - \sum_p W_p$. In particular the local distribution is

$$W_p(h) = L \sum_{k \geq 1} r^k \{h(kL) + h(-kL)\}, \quad h \in \mathcal{T}. \quad (2)$$

The sum is finite on each compactly supported test. The corresponding Hermitian forms are $B_p(f, g) = W_p(f * g^*)$ and $B_W(f, g) = W(f * g^*)$. The minus sign in the full Weil distribution follows from the classical explicit formula with its archimedean subtraction [4, Appendix B]. Consequently an intersection form intended to satisfy $B_W = -I$ has local increment $+B_p$, provided no further term changes at the same transition.

The subspace of tests with the two pole moments zero is

$$\mathcal{T}_0 = \{f \in \mathcal{T} : F_f(1/2) = F_f(-1/2) = 0\}. \quad (3)$$

No reality or evenness condition is imposed on $\mathcal{T}_0$. Our explicit witnesses will be real, which suffices

to prove both signs on this complex space. The pole term vanishes on $f * g^*$ when $f, g \in \mathcal{T}_0$; its vanishing does not remove the archimedean term or any finite prime.

For the analytic carrier, write $F_t = F(1/2 + it)$ and

$$q_S(F, G) = \int_{\mathbb{R}} F_t \overline{G_t} w_S(t) dt, \quad w_{S \cup \{p\}} = \omega_p w_S. \quad (4)$$

These are the common-carrier norms imported from Part I, Section 8, equations (46) and (47), and from equations (59) and (62) of [7]. The positive analytic form q_S and the hypothetical geometric form I_S are distinct objects. Equating either one with a desired arithmetic form is a comparison theorem to prove, not a convention.

All Hermitian forms in this paper are linear in the first argument. If $A : H_0 \rightarrow H_1$ is bounded, its adjoint satisfies $\langle Ax, y \rangle_1 = \langle x, A^*y \rangle_0$. Whenever an orthogonal projection onto an entire-function space is used below, that space is assumed to be a closed subspace of the ambient Hilbert realization. This hypothesis is essential for the projection calculation. The entire obstruction itself needs no Hilbert-space hypothesis.

3 Entire normalization

Definition 3.1. For a real number $p > 1$, the entire Euler denominator is

$$\Delta_p(z) = (1 - e^{-Lz})(1 - e^{L(z-1)}), \quad L = \log p. \quad (5)$$

The powers p^z always mean $e^{z \log p}$, where $\log p$ is the unique real number L satisfying $e^L = p$.

For $z = 1/2 + it$, the two factors in (5) are complex conjugates. Hence $\Delta_p(1/2 + it) = \omega_p(t) > 0$. Away from the critical line the same function has zeros, and their orders constrain any holomorphic square root.

Lemma 3.2. *The zeros of Δ_p are simple and occur at*

$$z = \frac{2\pi ik}{L} \quad \text{or} \quad z = 1 + \frac{2\pi ik}{L}, \quad k \in \mathbb{Z}. \quad (6)$$

In particular $\Delta_p(0) = 0$ and $\Delta'_p(0) = L(1 - p^{-1}) \neq 0$.

Proof. The equation $e^{-Lz} = 1$ is equivalent to the first lattice in (6); changing the sign of k does not change the set. Its factor has derivative $Le^{-Lz} = L$ at every zero. At such a point the other factor is $1 - p^{-1}$. The second factor vanishes at the second lattice and has derivative $-Le^{L(z-1)} = -L$ there. The first factor is then $1 - p^{-1}$. The two lattices have different real parts and cannot intersect. Thus exactly one factor vanishes at each zero, with nonzero derivative, which proves the claim. $\square$

Lemma 3.3. *Let F, G be entire functions. If*

$$G(1/2 + it) = \omega_p(t)^{-1/2} F(1/2 + it) \quad (7)$$

for Lebesgue-almost every $t \in \mathbb{R}$, then

$$\Delta_p(z) G(z)^2 = F(z)^2 \quad (z \in \mathbb{C}). \quad (8)$$

The same conclusion holds if the exceptional set is null for a measure $w(t) dt$ with $w > 0$ almost everywhere.

Proof. The difference of the two sides of (7) is continuous in t , because $\omega_p(t)$ is continuous and bounded below by $(1-r)^2 > 0$. A continuous function that is zero almost everywhere on $\mathbb{R}$ is zero everywhere: if its value were nonzero at t_0 , continuity would give an interval around t_0 on which its absolute value has a positive lower bound. That interval has positive measure. Squaring the resulting equality for every real t gives (8) on the critical line.

The function $\Delta_p G^2 - F^2$ is entire and vanishes, for example, at $1/2 + i/n$ for every positive integer n . These points accumulate at $1/2$ in its domain. The identity theorem makes the function zero on $\mathbb{C}$. If $w > 0$ almost everywhere, a set of positive Lebesgue measure has positive $w(t) dt$ measure: decompose it into the sets where $w \geq 1/n$. Thus weighted-almost-everywhere equality implies the equality used in the preceding argument. $\square$

Theorem 3.4 (Entire normalization obstruction). *For $p > 1$, there are no nonzero entire functions F, G satisfying (7) almost everywhere. In fact no nonzero entire functions satisfy (8).*

Proof. If a nonzero entire function vanishes to infinite order at zero, its Taylor series is zero in a neighborhood, and the identity theorem makes it identically zero. Hence nonzero entire F, G have finite orders $m = \text{ord}_0 F$ and $n = \text{ord}_0 G$ in $\mathbb{Z}_{\geq 0}$. Their local factorizations are

$$F(z) = z^m u(z), \quad G(z) = z^n v(z), \quad u(0)v(0) \neq 0.$$

Lemma 3.2 similarly gives $\Delta_p(z) = zd(z)$ with $d(0) \neq 0$. Equality (8) would therefore force $1 + 2n = 2m$. The left side is odd and the right side is even. This is impossible. Lemma 3.3 supplies (8) from the boundary relation. $\square$

The theorem applies to every nonzero element of an entire Sonin carrier, regardless of its zeros at 0 or 1. Adding finite-order zeros at the Euler lattice cannot remove an odd-order obstruction to a square. Allowing poles of finite order also does not help: the local orders then belong to $\mathbb{Z}$, and $1 + 2n = 2m$ remains impossible. A branch on a cut plane is a different analytic object and is outside the asserted entire realization.

Corollary 3.5. *Let $\mathcal{L}_S = L^2(\mathbb{R}, w_S dt)$ and $\mathcal{L}_T = L^2(\mathbb{R}, \omega_p w_S dt)$, where $w_S > 0$ almost everywhere. The polar isometry of the identity $j : \mathcal{L}_S \rightarrow \mathcal{L}_T$ is multiplication by $\omega_p^{-1/2}$. It does not restrict to a map from any nonzero entire-function carrier into itself.*

Proof. The bounds on ω_p make j a bounded bijection. Directly from the two inner products, j^* is multiplication by ω_p and $j^*j = M_{\omega_p}$. Functional calculus gives $j(j^*j)^{-1/2} = jM_{\omega_p^{-1/2}}$. The boundary output of a nonzero entire input cannot be the boundary of an entire function by Theorem 3.4. $\square$

3.1 Phases and divisors

The positive square root in polar decomposition is responsible for the parity argument. An arbitrary isometry between the ambient spaces may include a measurable phase $\chi(t)$ of modulus one: $F_t \mapsto \chi(t)\omega_p(t)^{-1/2}F_t$. Theorem 3.4 does not exclude every such phase. It excludes the positive pointwise polar factor of the specified carrier identity.

The complex reciprocal Euler factor illustrates the difference. Multiplication by $(1 - p^{-z})^{-1}$ has the same boundary modulus as $\omega_p^{-1/2}$, but its continuation is meromorphic. For an entire input F , the quotient $F/(1 - p^{-z})$ is entire exactly when F vanishes at every point $2\pi ik/L$. Necessity follows by comparing orders at each simple denominator zero. Sufficiency follows by removing those local singularities; the lattice has no finite accumulation point.

This cancellation condition is nonempty. For any entire H , the function $F = (1 - p^{-z})H$ has the entire quotient H . Whether both belong to a particular Sonin carrier requires that carrier's growth and norm conditions. The example only establishes the logical boundary of the positive-square-root obstruction.

There is a local version useful for other candidates. If a holomorphic function D has odd order at a point z_0 , then $DG^2 = F^2$ cannot hold for nonzero holomorphic germs F, G at z_0 . The preceding proof uses only local factorizations. Thus one does not need a global formula for the zeros of D once an odd zero has been found.

For a finite set of distinct primes E , the product $\Delta_E = \prod_{p \in E} \Delta_p$ has an odd zero whenever E is nonempty. If $|E|$ is odd, zero itself works. If $|E|$ is even, choose $p \in E$ and $z_0 = 2\pi i / \log p$. A second prime $q \neq p$ could have a zero there only if $\log q / \log p$ were a positive integer. This would give $q = p^k$, contrary to distinct primality. No second-lattice zero has real part zero. The zero of Δ_E at z_0 is therefore simple. The same argument excludes the positive pointwise normalization for every nonempty finite prime enlargement, including those with an even number of primes.

4 Compressed polar decomposition

Let $\mathcal{L} = L^2(\mathbb{R}, w dt)$, with $w > 0$ almost everywhere, and let $H \subset \mathcal{L}$ be closed. Suppose d is real and $0 < c \leq d \leq C < \infty$ almost everywhere. Write $q_0(x, y) = \langle x, y \rangle_{\mathcal{L}}$ and $q_1(x, y) = \langle M_d x, y \rangle_{\mathcal{L}}$ on H . Equivalent norms give the same closed vector subspace in the two ambient realizations. Write $H_i = (H, q_i)$ for $i = 0, 1$. The identity $J : H_0 \rightarrow H_1$ is a bounded bijection between these two Hilbert norms.

Proposition 4.1. *Let P be the q_0 -orthogonal projection of $\mathcal{L}$ onto H . The metric operator*

$$A = PM_d|_H \tag{9}$$

is positive, self-adjoint and boundedly invertible on H_0 , with $c \text{Id} \leq A \leq C \text{Id}$. The restricted identity has

$$J^* = A, \quad J^* J = A, \quad U = JA^{-1/2}. \tag{10}$$

Here $J^ : H_1 \rightarrow H_0$ uses the same vector formula A but different domain and target norms. The q_1 -orthogonal projection from the ambient space onto H is*

$$P_1 = A^{-1}PM_d. \tag{11}$$

Proof. For $x, y \in H$, orthogonality of P gives

$$q_1(x, y) = q_0(M_d x, y) = q_0(PM_d x, y) = q_0(Ax, y).$$

Since M_d is self-adjoint, this also equals $q_0(x, Ay)$. The bounds on d imply $c\|x\|_0^2 \leq q_0(Ax, x) \leq C\|x\|_0^2$. Thus A is bounded, self-adjoint and bounded below by c . Its range is closed, its kernel is zero, and its range is dense because its orthogonal complement is $\ker A^* = \ker A$. It is therefore surjective, with bounded inverse.

The displayed form identity gives the adjoint formula for J . The polar formula follows from the unique positive square root of A . In particular

$$q_1(A^{-1/2}x, A^{-1/2}y) = q_0(A^{1/2}x, A^{-1/2}y) = q_0(x, y).$$

Surjectivity of $A^{-1/2}$ makes U a unitary map between H_0 and H_1 .

For an ambient vector v , put $z = A^{-1}PM_d v \in H$. For every $h \in H$,

$$q_1(v - z, h) = q_0(PM_d v - Az, h) = 0.$$

This characterizes z as the q_1 -orthogonal projection of v . The operator is bounded in the ambient q_1 norm by equivalence of norms, proving (11). $\square$

Applied to $d = \omega_p$ and the closed entire Sonin realization, the proposition constructs an intrinsic polar isometry. Every output is an element of the same Hilbert space H , so it has the entire representative belonging to that realization. There is no conflict with Theorem 3.4: the expression $A^{-1/2}$ is not asserted to be $M_{\omega_p}^{-1/2}$.

If H is invariant under M_d , its orthogonal complement is also invariant because M_d is self-adjoint. Then P commutes with M_d , and continuous functional calculus identifies $A^{-1/2}$ with $M_{d^{-1/2}}|_H$. For $d = \omega_p$, a nonzero closed subspace consisting of entire boundary functions cannot have this invariance: its inverse square root would contradict the entire obstruction. Compression is therefore forced in any such nonzero realization.

4.1 A finite-dimensional model

Even a one-dimensional subspace separates compressed and pointwise normalization. Take $\mathcal{L} = \mathbb{C}^2$, $H = \text{span}\{(1, 1)\}$ and $d = \text{diag}(1, 4)$. For $x = (a, a)$ one has $q_0(x, x) = 2|a|^2$ and $q_1(x, x) = 5|a|^2$. The metric operator on H is $A = (5/2)\text{Id}$. Intrinsic normalization sends (a, a) to $\sqrt{2/5}(a, a)$. Ambient normalization sends it to $(a, a/2)$, which is outside H unless $a = 0$. Both operations are isometries on their stated domains.

This example also shows why replacing functional calculus of a compression by compression of functional calculus is invalid. On H , the operator $PM_{d^{-1/2}}|_H$ is $(3/4)\text{Id}$, whereas $A^{-1/2} = \sqrt{2/5}\text{Id}$. Their scalar values differ. No approximation or infinite-dimensional issue is involved.

4.2 Test maps and symmetry

An arithmetic test map specifies which geometric vector represents each compact smooth test in the comparison formula. Changing that vector changes the pairing being compared with the Weil form. Suppose $\Gamma_0 : \mathcal{T} \rightarrow H_0$ and $\Gamma_1 : \mathcal{T} \rightarrow H_1$ are complex-linear maps satisfying the original compatibility $\Gamma_1 = J\Gamma_0$. Replacing J by U while retaining both test maps requires

$$A^{-1/2}\Gamma_0 f = \Gamma_0 f \quad (f \in \mathcal{T}). \quad (12)$$

Proposition 4.2. *If $\Gamma_0(\mathcal{T})$ is dense in H_0 , the original and polar test-map compatibilities both hold exactly when $A = \text{Id}$. Without density they require A to be the identity on the closure of $\Gamma_0(\mathcal{T})$.*

Proof. The difference $A^{-1/2} - \text{Id}$ is bounded. If it vanishes on $\Gamma_0(\mathcal{T})$, it vanishes on its closure. On that closure, $A^{-1/2}x = x$ implies $A^{1/2}x = x$ by applying $A^{1/2}$, and then $Ax = x$. Conversely $Ax = x$ implies $A^{-1/2}x = x$ by functional calculus, or by uniqueness of the positive square root on the eigenspace for eigenvalue one. The dense case makes this identity hold on all of H_0 . $\square$

The proposition is a conditional test. It does not claim that the compressed Sonin metric operator is nontrivial on any particular arithmetic test image. A proof of compatibility must either establish (12), change the target test map, or supply a new comparison theorem for the changed image. Redefining $\Gamma_1 = U\Gamma_0$ proves an analytic identity by construction but does not preserve an independently defined arithmetic correspondence automatically.

Fourier compatibility can survive compression under verifiable symmetry hypotheses. Let R be a unitary or antiunitary operator on $\mathcal{L}$ that maps H onto H and commutes with M_d . Since R maps $H^\perp$ onto itself, it commutes with P . Therefore $RA = AR$ on H . Real polynomial approximation to $x^{-1/2}$ on $[c, C]$ gives $RA^{-1/2} = A^{-1/2}R$; real coefficients also cover the antiunitary case.

For example, reflection $v(t) \mapsto v(-t)$ commutes with M_{ω_p} because ω_p is even. It is unitary when the ambient weight is even. If the carrier is reflection invariant, the intrinsic normalization respects that symmetry. To call this the Fourier duality of a semilocal model, one must identify its actual Fourier operator in the chosen carrier. The shell transition of Part I has such a published intertwining identity. The conditional reflection calculation alone does not identify every geometric or arithmetic duality operator.

5 Compatibility after compression

The scalar Euler multipliers for distinct primes commute. Compression does not preserve this implication for their metric operators. Let M_a, M_b be commuting self-adjoint ambient multipliers and P an orthogonal projection. On H , put $A = PM_a|_H$ and $B = PM_b|_H$. Expanding $\text{Id} = P + (\text{Id} - P)$ between the ambient multipliers gives

$$AB - BA = PM_b(\text{Id} - P)M_a|_H - PM_a(\text{Id} - P)M_b|_H. \quad (13)$$

The right side records the two excursions outside H . Commutativity of M_a, M_b does not force either term to vanish.

An explicit example uses the orthonormal vectors

$$u = \frac{(1, -1, 0)}{\sqrt{2}}, \quad v = \frac{(1, 1, -2)}{\sqrt{6}}$$

in $\mathbb{C}^3$ and $H = \text{span}\{u, v\}$. Compress the two coordinate projections $E_1 = \text{diag}(1, 0, 0)$ and $E_2 = \text{diag}(0, 1, 0)$. In the basis (u, v) the compressed matrices are

$$A_1 = \begin{pmatrix} 1/2 & 1/(2\sqrt{3}) \\ 1/(2\sqrt{3}) & 1/6 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1/2 & -1/(2\sqrt{3}) \\ -1/(2\sqrt{3}) & 1/6 \end{pmatrix}. \quad (14)$$

Multiplying the matrices yields

$$A_1 A_2 - A_2 A_1 = \begin{pmatrix} 0 & -1/(3\sqrt{3}) \\ 1/(3\sqrt{3}) & 0 \end{pmatrix} \neq 0. \quad (15)$$

Replacing the ambient projections by $\text{Id} + E_1$ and $\text{Id} + E_2$ makes them positive and invertible without changing the compressed commutator. These positive diagonal operators are finite-dimensional models of commuting ambient multiplication operators; their compressed matrices show the possible failure of commutation. Thus strict positivity does not repair the failure.

The example is a statement about compression, not a calculation of the actual Sonin projection. It prevents an argument that uses scalar commutativity as its only justification for commuting compressed metric operators. The Sonin operators may have additional identities; those identities need proof.

The second metric transition must use the inner product produced by the first. Express both intermediate and final forms through the base form by setting $q_p(x, y) = q_0(C_p x, y)$ and $q_{pq}(x, y) = q_0(C_{pq} x, y)$, where

$$C_p = PM_{\omega_p}|_H, \quad C_{pq} = PM_{\omega_p \omega_q}|_H.$$

The metric operator from q_p to q_{pq} is

$$A_{p,q} = C_p^{-1} C_{pq}, \quad (16)$$

as an operator on the intermediate Hilbert space (H, q_p) . Its positivity and adjoint are determined by q_p , as the identity $q_p(A_{p,q}x, y) = q_0(C_{pq}x, y) = q_{pq}(x, y)$ shows. It need not be self-adjoint for the base form q_0 . Its positive square root must therefore be taken in (H, q_p) .

Writing every underlying carrier map as the identity, the polar path through p acts by $A_{p,q}^{-1/2} C_p^{-1/2}$, and the path through q acts by $A_{q,p}^{-1/2} C_q^{-1/2}$. A commuting polar square therefore requires equality of these two operators. The calculation above neither proves nor disproves that identity for Sonin spaces. It identifies the operator equation that replaces commutation of the scalar Euler factors.

There is always an abstract way to choose coherent isometries relative to one base metric. If $q_S(x, y) = q_0(C_S x, y)$ on a common Hilbert carrier, put $V_S = C_S^{-1/2} : H_0 \rightarrow H_S$ and $V_{S,T} = V_T V_S^{-1}$. Then $V_{T,U} V_{S,T} = V_{S,U}$ by cancellation. These maps are isometries and form commuting squares. They depend on the chosen base metric and need not be the polar factors of the one-step carrier identities. Their arithmetic test-map compatibility is again a separate equation. Abstract coherence does not construct a local arithmetic correspondence.

6 Smooth local witnesses

Fix $p > 1$, $L = \log p$ and $r = p^{-1/2}$. Choose $0 < \epsilon < L/2$ and a nonzero real function $\psi \in C_c^\infty(\mathbb{R})$ supported in $[-\epsilon, \epsilon]$. For instance one may use

$$\psi(x) = \begin{cases} \exp(-1/(1 - (x/\epsilon)^2)), & |x| < \epsilon, \\ 0, & |x| \geq \epsilon. \end{cases} \quad (17)$$

All derivatives tend to zero at the endpoints because every polynomial in $(1 - (x/\epsilon)^2)^{-1}$ is dominated by its negative exponential. Thus this is a compact smooth function, positive near zero, and $\|\psi\|_2^2 > 0$. No unit-mass normalization is required: every local quadratic value below retains the same positive factor $\|\psi\|_2^2$.

For a finite complex vector $c = (c_0, \dots, c_n)$, write

$$f_c(x) = \sum_{j=0}^n c_j \psi(x - jL), \quad C_c(z) = \sum_{j=0}^n c_j z^j. \quad (18)$$

The translates have disjoint supports. Hence

$$\|f_c\|_2^2 = \|\psi\|_2^2 \sum_{j=0}^n |c_j|^2. \quad (19)$$

In particular a nonzero coefficient vector produces a nonzero test, without any possible cancellation between adjacent bumps.

Proposition 6.1. *For the packet (18),*

$$F_{f_c}(z) = F_\psi(z) C_c(e^{Lz}), \quad (20)$$

$$W_p(f_c * f_c^*) = L \|\psi\|_2^2 Q_r(c), \quad (21)$$

$$Q_r(c) = \sum_{i \neq j} c_i \bar{c}_j r^{|i-j|}. \quad (22)$$

Consequently $C_c(r) = C_c(r^{-1}) = 0$ implies $f_c \in \mathcal{T}_0$.

Proof. Changing variables $u = x - jL$ in the integral for the Laplace transform gives $F_{\psi(\cdot - jL)}(z) = e^{jLz} F_{\psi}(z)$. Summing the finitely many terms proves (20) for every complex z . At $z = 1/2$ and $z = -1/2$ the polynomial arguments are r^{-1} and r , respectively.

Put $h_{\psi} = \psi * \psi^*$. Its support lies in $[-2\epsilon, 2\epsilon]$ and $h_{\psi}(0) = \|\psi\|_2^2$. Expanding the finite convolution sum gives

$$(f_c * f_c^*)(x) = \sum_{i,j=0}^n c_i \bar{c}_j h_{\psi}(x - (i - j)L).$$

At $x = kL$ with integer k , every term vanishes unless $k = i - j$, because $2\epsilon < L$. Therefore, for $k \geq 1$,

$$\begin{aligned} (f_c * f_c^*)(kL) &= \|\psi\|_2^2 \sum_{i-j=k} c_i \bar{c}_j, \\ (f_c * f_c^*)(-kL) &= \|\psi\|_2^2 \sum_{j-i=k} c_i \bar{c}_j. \end{aligned}$$

These sums are zero for $k > n$. Substitution in the actual local distribution (2) now gives (21). Each ordered pair $i \neq j$ occurs once, with its weight $r^{|i-j|}$. This also shows directly that $Q_r(c)$ is real, since reversing i, j conjugates a term. $\square$

The condition on the support of ψ is exact. There is no limit as ϵ tends to zero, and the normalization $\|\psi\|_2^2$ is strictly positive. All prime powers that can contribute to the packet are included in (21). Terms at larger powers vanish by support, rather than being discarded as small.

Let $a_r = r + r^{-1}$ and consider

$$P_r(z) = 1 - a_r z + z^2 = (z - r)(z - r^{-1}). \quad (23)$$

Any coefficient polynomial divisible by P_r has both required roots. This provides a direct way to impose the pole conditions before testing the sign, without projecting a numerical eigenvector onto an approximate moment kernel.

Lemma 6.2. For $0 < r < 1$ and $a_r = r + r^{-1}$,

$$Q_r(1, -a_r, 1) = -4 - 2r^2, \quad (24)$$

$$Q_r(1, 1 - a_r, 1 - a_r, 1) = -8 - 4r^2 + 6r + 2/r. \quad (25)$$

The respective coefficient polynomials are $P_r(z)$ and $P_r(z)(1 + z)$.

Proof. For real coefficients one can sum by distance:

$$Q_r(c) = 2 \sum_{k=1}^n r^k \sum_{j=0}^{n-k} c_{j+k} c_j.$$

For the three-term vector the inner sums are $-2a_r$ at distance one and 1 at distance two. Thus $Q = -4a_r r + 2r^2 = -4 - 2r^2$.

For the four-term vector set $b = 1 - a_r$. The inner sums are $2b + b^2$ at distance one, $2b$ at distance two, and 1 at distance three. Hence

$$\begin{aligned} Q &= 2r(2b + b^2) + 4r^2 b + 2r^3 \\ &= 2r(3 - 4a_r + a_r^2) + 4r^2(1 - a_r) + 2r^3. \end{aligned}$$

Using $a_r r = 1 + r^2$ and $a_r^2 = r^2 + 2 + r^{-2}$ reduces this expression to (25). Multiplication of P_r by $1 + z$ gives coefficients $(1, 1 - a_r, 1 - a_r, 1)$, which proves the polynomial statements. $\square$

Theorem 6.3 (Exact signs at eleven). *There exist nonzero real tests $f_-, f_+ \in \mathcal{T}_0$ such that*

$$W_{11}(f_- * f_-^*) = -\frac{46}{11}(\log 11)\|\psi\|_2^2 < 0, \quad (26)$$

$$W_{11}(f_+ * f_+^*) = \frac{28\sqrt{11} - 92}{11}(\log 11)\|\psi\|_2^2 > 0. \quad (27)$$

They can be chosen as three and four translates of any bump (17) with $2\epsilon < \log 11$.

Proof. Set $r = 1/\sqrt{11}$, so $a_r = 12/\sqrt{11}$, and use the two coefficient vectors in Lemma 6.2. Their polynomials are divisible by P_r , so Proposition 6.1 gives both pole moments exactly zero. Their first coefficient is one, and (19) makes both tests nonzero.

Substituting $r^2 = 1/11$ into (24) gives $-46/11$. Substituting $r = 1/\sqrt{11}$ into (25) gives $(28\sqrt{11} - 92)/11$. This is positive because $7\sqrt{11} > 23$: both sides are positive and their squares satisfy $49 \cdot 11 = 539 > 529 = 23^2$. Finally $\log 11 > 0$ and $\|\psi\|_2^2 > 0$, so the signs are unchanged in the smooth distribution values. $\square$

Translation of the complete packet by a common real number does not change its autocorrelation. It multiplies each Laplace moment by a nonzero exponential and therefore preserves a zero moment. The negative packet can consequently be centered in $[-L - \epsilon, L + \epsilon]$, and the positive packet in $[-3L/2 - \epsilon, 3L/2 + \epsilon]$. This centering will make the support condition in the geometric obstruction explicit.

7 Finite packets at every prime

The short positive packet at 11 is sufficient for the orthogonal-extension obstruction. A longer packet gives the same two-sign statement at every prime. Its exact formula also explains why moment cancellation does not restore the sign of a local distribution.

Proposition 7.1. *For $0 < r < 1$, $N \geq 2$, let $c^{(N)}$ be the coefficient vector of*

$$C_N(z) = P_r(z)(1 + z + \cdots + z^{N-1}). \quad (28)$$

Then

$$Q_r(c^{(N)}) = \frac{2N(1-r)^3}{r} + 4 - 6r - \frac{2}{r}. \quad (29)$$

Proof. On the unit circle $z = e^{i\theta}$ the absolutely convergent Fourier series of the off-diagonal Toeplitz kernel is

$$k_r(\theta) = \sum_{k \neq 0} r^{|k|} e^{ik\theta} = \frac{1 - r^2}{1 - r(z + z^{-1}) + r^2} - 1. \quad (30)$$

Expanding a finite polynomial C and integrating its product with this series term by term gives

$$Q_r(c) = \frac{1}{2\pi} \int_{-\pi}^{\pi} k_r(\theta) |C(e^{i\theta})|^2 d\theta. \quad (31)$$

Absolute convergence justifies the interchange, and orthogonality of integer exponentials leaves exactly the coefficient sum in (22).

Put $s = z + z^{-1}$ and $D = 1 - rs + r^2$. The moment polynomial in (23) satisfies

$$P_r(z) = z(s - r - r^{-1}) = -zD/r.$$

For $|z| = 1$, D is real, so $|P_r(z)|^2 = D^2/r^2$. Multiplying (30) by this square cancels the denominator:

$$k_r(\theta)|P_r(z)|^2 = \frac{(rs - 2r^2)D}{r^2} \quad (32)$$

$$= (-4 - 2r^2) + (3r + r^{-1})(z + z^{-1}) - (z^2 + z^{-2}). \quad (33)$$

For completeness, expansion of the first line yields $(r^{-1} + 3r)s - s^2 - 2 - 2r^2$, and $s^2 = z^2 + 2 + z^{-2}$ gives the second line.

The polynomial $B_N(z) = 1 + \dots + z^{N-1}$ satisfies

$$|B_N(z)|^2 = \sum_{k=-(N-1)}^{N-1} (N - |k|)z^k.$$

The constant coefficient of $k_r|P_r|^2|B_N|^2$ is therefore

$$N(-4 - 2r^2) + 2(N - 1)(3r + r^{-1}) - 2(N - 2).$$

This formula includes $N = 2$, when the final contribution is zero. Simplifying its coefficient of N gives $2(r^{-1} - 3 + 3r - r^2) = 2(1 - r)^3/r$, and its constant term is $4 - 6r - 2/r$. Equation (31) proves (29). $\square$

Corollary 7.2. *For every prime p , the quadratic form $f \mapsto W_p(f * f^*)$ has both positive and negative values on nonzero real tests in $\mathcal{T}_0$.*

Proof. The three-term packet always gives $-4 - 2r^2 < 0$. The coefficient of N in (29) is strictly positive for $0 < r < 1$. Choose an integer $N \geq 2$ such that

$$N > \frac{r(6r + 2/r - 4)}{2(1 - r)^3}.$$

Then (29) is positive. The polynomial C_N is divisible by P_r and has first coefficient one, so its smooth packet is nonzero and belongs to $\mathcal{T}_0$. Proposition 6.1 transfers the exact positive coefficient value to W_p . $\square$

The support of the longer positive packet grows with N . The corollary therefore gives a local sign result on the union of compact test supports. It gives no common small support on which every prime is visible. On a fixed support, all sufficiently large primes have identically zero local form. This restriction is part of the arithmetic geometry of the test system, rather than a defect in the packet construction.

The Toeplitz symbol itself is positive at $\theta = 0$ and negative at $\theta = \pi$. That observation would suggest frequency localization, but it would not by itself impose the two moment conditions. The finite factor P_r imposes those conditions exactly. Its roots r, r^{-1} are off the unit circle, so the factor leaves nontrivial Fourier packets available. The cancellation in (33) turns that observation into the finite proof above.

8 The constrained finite form

The four-bump example at 11 has a useful minimality property within this specific translated-bump family. The two moment conditions reduce the coefficient space to a polynomial divisibility condition, and the restricted quadratic form can be diagonalized exactly. This is a finite-dimensional statement about equally spaced translates of the same bump. It is not a minimal-support theorem for arbitrary compact smooth tests.

Proposition 8.1. *Let $0 < r < 1$, and let $c = (c_0, c_1, c_2, c_3) \in \mathbb{C}^4$ satisfy $C_c(r) = C_c(r^{-1}) = 0$. There are unique $\alpha, \beta \in \mathbb{C}$ such that*

$$C_c(z) = P_r(z)(\alpha + \beta z). \quad (34)$$

On these coordinates the constrained form is

$$Q_r(c) = b_0(|\alpha|^2 + |\beta|^2) + 2b_1 \operatorname{Re}(\alpha\bar{\beta}), \quad b_0 = -4 - 2r^2, \quad b_1 = 3r + r^{-1}. \quad (35)$$

Its eigenvalues in the coefficient coordinates (α, β) are $b_0 + b_1$ and $b_0 - b_1$.

Proof. The roots r and r^{-1} are distinct, so their two linear factors divide C_c . Since C_c has degree at most three and P_r is monic of degree two, the quotient has degree at most one and is unique. This proves (34), including the zero polynomial. Substitute $C_c = P_r(\alpha + \beta z)$ into (31) and use (33). The product

$$|\alpha + \beta z|^2 = |\alpha|^2 + |\beta|^2 + \bar{\alpha}\beta z + \alpha\bar{\beta}z^{-1}$$

has frequencies only zero and one. Its constant coefficient after multiplication by the Laurent polynomial is the right side of (35); the terms at frequency two contribute nothing. The matrix is $\begin{pmatrix} b_0 & b_1 \\ b_1 & b_0 \end{pmatrix}$. Its eigenvectors $(1, 1)$ and $(1, -1)$ have the stated eigenvalues. $\square$

The word eigenvalue here refers to the Euclidean norm of the quotient coefficients (α, β) . The original four coefficients have a different induced norm under (34). The signature, which is what the sign test uses, is unchanged by this injective change of coordinates.

Corollary 8.2. *Among arithmetic primes, 11 is the smallest prime for which four equally spaced bumps can give a positive local value while their coefficient polynomial has both moment roots. For $p \leq 7$ the form on that two-dimensional coefficient kernel is negative definite. For every prime $p \geq 11$ its signature is $(1, 1)$.*

Proof. The eigenvalue $b_0 - b_1$ is negative for every $0 < r < 1$. The sign of $b_0 + b_1$ is the sign of

$$g(r) = r(b_0 + b_1) = 1 - 4r + 3r^2 - 2r^3.$$

Its derivative is $g'(r) = -4 + 6r - 6r^2 = -6(r - 1/2)^2 - 5/2 < 0$. At $r = 1/\sqrt{11}$, the value of $b_0 + b_1$ is $(14\sqrt{11} - 46)/11 > 0$, by the integer-square comparison used in Theorem 6.3. It remains positive for smaller r , which includes every prime $p \geq 11$.

At $r = 1/\sqrt{7}$,

$$g(r) = \frac{10\sqrt{7} - 30}{7\sqrt{7}} < 0,$$

because $\sqrt{7} < 3$. It remains negative for larger r , which includes every prime at most seven. The only primes below eleven are 2, 3, 5, 7; the intervening integers 8, 9, 10 are composite. This proves all the assertions. $\square$

Three equally spaced bumps cannot give a positive value under the same coefficient moment conditions, at any prime. Their polynomial has degree at most two, so it must be αP_r . Its quadratic value is $|\alpha|^2(-4 - 2r^2)$, strictly negative unless the test is zero. With two bumps or one bump, a polynomial of degree at most one with both distinct roots is zero. Thus the three-term negative witness and the four-term positive witness at 11 use the smallest possible numbers of translates within this coefficient construction.

For the explicit nonnegative bump (17), both values $F_\psi(1/2)$ and $F_\psi(-1/2)$ are strictly positive. Equation (20) then makes the coefficient root conditions equivalent to the actual two zero moments. The minimality statement applies to all packets built from this bump, rather than just to a sufficient algebraic condition. If a different bump already has zero moments, the equivalence can fail, and this argument does not give the same minimality statement for that family.

9 Orthogonal primitive extensions

Let $\mathcal{D}$ be a complex test space on which B_p is defined. It may be $\mathcal{T}_0$ or a subspace containing the specific witness under consideration. Let P_S, P_T be complex vector spaces equipped with Hermitian forms I_S, I_T . No completeness assumption is needed in the following algebraic argument.

Definition 9.1. An orthogonal nonpositive extension on $\mathcal{D}$ consists of an isometric complex-linear inclusion $\iota : P_S \rightarrow P_T$ and complex-linear maps $\Gamma_S : \mathcal{D} \rightarrow P_S, \Gamma_T, \eta : \mathcal{D} \rightarrow P_T$ such that

$$\Gamma_T f = \iota \Gamma_S f + \eta f, \quad (36)$$

$$I_T(\iota x, \eta f) = 0 \quad (x \in P_S, f \in \mathcal{D}), \quad (37)$$

$$I_T(\eta f, \eta f) \leq 0 \quad (f \in \mathcal{D}). \quad (38)$$

The isometry condition means $I_T(\iota x, \iota y) = I_S(x, y)$.

Nonpositivity of all of I_T implies (38), but the definition only needs the sign on the added test image. Thus the exclusion below also applies when the ambient geometric pairing has other signs outside the relevant sector.

Theorem 9.2 (Orthogonal extension obstruction). *Suppose $\mathcal{D}$ contains the positive test f_+ in Theorem 6.3. There is no orthogonal nonpositive extension on $\mathcal{D}$ for which the one-prime comparison at $p = 11$ is*

$$I_T(\Gamma_T f, \Gamma_T g) - I_S(\Gamma_S f, \Gamma_S g) = W_{11}(f * g^*) \quad (f, g \in \mathcal{D}). \quad (39)$$

For an arbitrary prime p , the same conclusion holds on $\mathcal{D} = \mathcal{T}_0$ with W_p in place of W_{11} .

Proof. Expand (36) in the Hermitian form. The old-old term cancels by isometry. The two mixed terms are zero by (37) and Hermitian symmetry. Therefore

$$I_T(\Gamma_T f, \Gamma_T f) - I_S(\Gamma_S f, \Gamma_S f) = I_T(\eta f, \eta f) \leq 0.$$

At $f = f_+$ this contradicts the strict positivity in (27). For every prime, use the positive test supplied by Corollary 7.2. $\square$

The proof tests a prescribed equation and sign together. It does not assume an intersection form exists and then derive a contradiction from the existence alone. Removing orthogonality, allowing corrections in the comparison, or requiring sign only at a later support-complete stage changes the hypotheses of the theorem.

The opposite local sign is useful as well. If a proposed orthogonal sector were nonnegative and its prescribed increment were $+W_p$, the negative packet would exclude that version. If the intended comparison used $-W_p$, the roles of the two packets would interchange. The sign issue is not repaired by choosing one uniform sign for an orthogonal local summand on all tests in $\mathcal{T}_0$.

9.1 Mixed pairings

Proposition 9.3. *Suppose $\iota : P_S \rightarrow P_T$ is isometric and $\Gamma_T = \iota\Gamma_S + \eta$, without orthogonality. Then the diagonal increment is*

$$I_T(\Gamma_T f, \Gamma_T f) - I_S(\Gamma_S f, \Gamma_S f) = 2 \operatorname{Re} I_T(\iota\Gamma_S f, \eta f) + I_T(\eta f, \eta f). \quad (40)$$

If isometry is also dropped, one must add

$$E(f, g) = I_T(\iota\Gamma_S f, \iota\Gamma_S g) - I_S(\Gamma_S f, \Gamma_S g) \quad (41)$$

to the corresponding sesquilinear expansion.

Proof. Sesquilinearity expands the pairing of $\iota\Gamma_S f + \eta f$ with itself into four terms. The two mixed terms are complex conjugates, giving the real part in (40). The old-old term cancels under isometry. Without isometry its remainder is exactly (41). The same expansion for two different tests gives the sesquilinear statement. $\square$

Even a negative definite one-dimensional form permits a positive increment through mixed terms. On $\mathbb{C}$, take $I(x, y) = -x\bar{y}$, old vector $x = 1$ and added vector $y = -1/2$. Then

$$I(x + y, x + y) - I(x, x) = -1/4 + 1 = 3/4 > 0,$$

although $I(y, y) = -1/4 < 0$. Formula (40) gives a mixed contribution of 1. This explicit example rules out any inference that a nonpositive total pairing makes every transition increment nonpositive.

There is a quantitative constraint if the whole form is nonpositive. Applying Cauchy-Schwarz to the positive semidefinite form $-I_T$ gives

$$|I_T(x, y)|^2 \leq (-I_T(x, x))(-I_T(y, y)).$$

For completeness, this inequality follows by requiring $-I_T(x + zy, x + zy) \geq 0$ for all complex z and minimizing the resulting quadratic polynomial; if $I_T(y, y) = 0$, the linear term must vanish. Thus mixed terms must obey a bound, but they can have either sign and can dominate the added diagonal term. A geometric construction must calculate them before applying any sign argument.

10 Support-indexed comparison

Part I, Definition 12.1, uses pairs (S, a) with all primes at most e^{2a} included in S . Let $\mathcal{T}_a$ denote its tests supported in $[-a, a]$. The following locality statement explains exactly when the orthogonal extension obstruction can be applied.

Proposition 10.1. *If $f, g \in \mathcal{T}_a$ and $p > e^{2a}$, then $W_p(f * g^*) = 0$. Consequently, if (S, a) is support admissible and $p \notin S$, adjoining p has zero local Weil increment on $\mathcal{T}_a$.*

Proof. If $(f * g^*)(x) \neq 0$, there are points u and v in the supports of f and g with $x = u - v$. Thus $\operatorname{supp}(f * g^*) \subset [-2a, 2a]$. If $p > e^{2a}$, then $k \log p > 2a$ for every $k \geq 1$. Both evaluations in (2) vanish. At a support-admissible pair, every prime outside S satisfies this strict inequality. $\square$

For the centered positive witness at 11, one may take $a = 3 \log 11/2 + \epsilon$. A support-complete place set therefore contains all primes at most $11^3 e^{2\epsilon}$, including 11. The transition that adds 11 while testing this function starts from an incomplete set for this support. Theorem 9.2 concerns

the stronger demand that this intermediate transition itself have an orthogonal nonpositive added image and exact increment.

In a support-indexed construction, an admissible enlargement from (S, a) to (T, b) with $b > a$ admits new tests. A prime already irrelevant at radius a may contribute at radius b . The comparison on the old tests and the construction on the new tests are different obligations. Fixing a test and telescoping local distributions across the primes in $T \setminus S$ is valid algebraically, but intermediate geometric sectors need not satisfy the final sign theorem.

At the level of the local distributions, path independence is exact. For finite disjoint sets of primes E, F ,

$$\sum_{p \in E \cup F} W_p(h) = \sum_{p \in E} W_p(h) + \sum_{p \in F} W_p(h).$$

This identity follows from a finite sum. If geometric transition defects are to reproduce it, their mixed and boundary terms must telescope to the same result. It does not follow that each intermediate diagonal defect has a fixed sign.

10.1 A finite support bound

There is a simple continuity estimate for each local form. For $f, g \in \mathcal{T}_a$, Cauchy-Schwarz applied to their correlation gives $|(f * g^*)(x)| \leq \|f\|_2 \|g\|_2$ for every x . Let $M = \lfloor 2a/L \rfloor$. Then

$$|W_p(f * g^*)| \leq 2L \left(\sum_{k=1}^M r^k \right) \|f\|_2 \|g\|_2 \leq \frac{2Lr}{1-r} \|f\|_2 \|g\|_2. \quad (42)$$

The empty sum is zero. The estimate extends the local form continuously to L^2 tests supported in the same interval. It also proves continuity on every compact test stage.

Only finitely many primes contribute on $\mathcal{T}_a$, so the sum of their local forms is continuous on that stage. This does not provide a uniform bound as a grows, nor does it address the singular archimedean subtraction. Those two issues belong to completion of the full comparison. The local sign witnesses and their obstruction require neither an infinite Euler product nor a global norm limit.

11 Pole moments and completion

Removing the two pole moments is a useful way to test a primitive ansatz, but it does not give a dense subspace from which the full sign follows by continuity. The moment functionals are continuous in the compact smooth topology. They are also continuous in $L^2([-a, a])$, since

$$|F_f(\pm 1/2)| \leq \|f\|_2 \left(\int_{-a}^a e^{\pm x} dx \right)^{1/2}.$$

Their common kernel is therefore closed in either setting.

Proposition 11.1. *On $\mathcal{T}$, the two functionals $f \mapsto F_f(1/2)$ and $f \mapsto F_f(-1/2)$ are linearly independent and have a two-dimensional smooth complement. For any $a > 0$ the same statement holds on $\mathcal{T}_a$.*

Proof. Choose a nonnegative nonzero compact smooth bump β whose support and a nonzero translate by b are both contained in $(-a, a)$. This is possible by first taking the support sufficiently

small. Let $\beta_0 = \beta$ and $\beta_1(x) = \beta(x - b)$. Set $u = F_\beta(1/2) > 0$ and $v = F_\beta(-1/2) > 0$. The matrix of their two moments is

$$M = \begin{pmatrix} u & e^{b/2}u \\ v & e^{-b/2}v \end{pmatrix}. \quad (43)$$

Its determinant is $uv(e^{-b/2} - e^{b/2}) \neq 0$ because $b \neq 0$. Thus the moment map is onto $\mathbb{C}^2$ on $\text{span}\{\beta_0, \beta_1\}$. For each f , solve

$$M \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} F_f(1/2) \\ F_f(-1/2) \end{pmatrix}.$$

Then $f - c_0\beta_0 - c_1\beta_1$ has both moments zero, and the decomposition is unique because M is invertible. All functions remain in $\mathcal{T}_a$. The same construction works in $\mathcal{T}$ without a fixed support bound. $\square$

Suppose a Hermitian form B has a known sign on $\mathcal{T}_0$. The decomposition from Proposition 11.1 gives a block form on $\mathcal{T}_0 \oplus \text{span}\{\beta_0, \beta_1\}$. A sign on the first block leaves the finite degree block and its mixed entries undetermined. For $f = f_0 + v$ the exact identity is

$$B(f, f) = B(f_0, f_0) + 2 \operatorname{Re} B(f_0, v) + B(v, v).$$

This is why a comparison proved only after killing moments needs a separate treatment of degree and pole corrections before the all-test Weil criterion can be invoked.

11.1 Limits of finite norm comparisons

For a finite prime set E , the local norm bounds multiply:

$$\prod_{p \in E} (1 - p^{-1/2})^2 q_S(F, F) \leq q_{S \cup E}(F, F) \leq \prod_{p \in E} (1 + p^{-1/2})^2 q_S(F, F). \quad (44)$$

These are finite inequalities. Without a separate estimate they do not give constants uniform in E . An arithmetic completion might use another topology, renormalized pairings or a support-indexed limit. The needed estimate must be proved in that stated topology.

A related operator distinction can be established abstractly. Let H be a Hilbert space, P_n proper orthogonal projections, and $T_n = P_n A_n P_n$ bounded operators on H . If T is bounded below by $c > 0$, then

$$\|T - T_n\| \geq c \quad \text{for every } n. \quad (45)$$

Indeed choose a unit vector $v_n \in \ker P_n$. Then $T_n v_n = 0$, while $\|T v_n\| \geq c$. Thus zero-extended proper cutoffs cannot converge in operator norm to a bounded-below operator.

Strong convergence can still hold. Suppose T_n are uniformly bounded by M and $T_n x$ converges for every x in a dense linear subspace $D \subset H$. For arbitrary $x \in H$, choose $y \in D$ with $\|x - y\| < \epsilon/(4M)$ when $M > 0$. For sufficiently large n, m ,

$$\|T_n x - T_m x\| \leq 2M\|x - y\| + \|T_n y - T_m y\| < \epsilon.$$

Completeness gives $Tx = \lim_n T_n x$; the limit is linear and bounded by M . The case $M = 0$ is trivial. This argument specifies sufficient hypotheses for a strong completion, but neither hypothesis follows merely from the one-prime bounds in (44).

11.2 Dynamics and domains

The usual Sonin condition imposes simultaneous spatial and Fourier gaps. With the Fourier convention $\mathcal{F}f(\xi) = \int f(x)e^{-2\pi i x \xi} dx$, the unitary dilation $D_b f(x) = b^{1/2} f(bx)$ satisfies

$$\mathcal{F}(D_b f)(\xi) = b^{-1/2} \mathcal{F}f(\xi/b).$$

If both original gaps have radius λ , the transformed spatial gap has radius λ/b and the Fourier gap has radius $b\lambda$. These formulae follow by substitution in the support conditions and in the Fourier integral. They do not establish invariance of a fixed Sonin space under the full dilation group.

A weight-one adjoint identity also has a domain condition. If a densely defined operator Θ satisfies $\text{Dom } \Theta^* = \text{Dom } \Theta$ and $\Theta^* = \text{Id} - \Theta$ on that domain, then $A = (\Theta - \frac{1}{2} \text{Id})/i$ is self-adjoint. To check this, set $B = \Theta - \frac{1}{2} \text{Id}$. The bounded shift leaves the adjoint domain unchanged and gives $B^* = -B$. Since $A = -iB$, the scalar adjoint rule yields $A^* = iB^* = -iB = A$ on the same domain. An equality of differential expressions on a test core does not imply this domain equality.

This conditional operator statement does not supply a realization of the zeta zeros. A spectral argument would also have to identify every nontrivial zero, with its multiplicity, in the stated operator realization. The entire normalization and local sign arguments above do not make that identification.

12 Arithmetic geometry and prior work

The geometric motivation has several distinct mathematical sources. Semilocal adelic quotients and their crossed-product algebras provide arithmetic actions and local trace formulae. Connes's semilocal cutoff theorem retains a divergent term and Fourier-normalized principal values [2, Section VII, Theorem 4]. His global trace condition in the following section has an RH-equivalent status. The finite trace theorem therefore cannot be used as an unconditional global positivity theorem.

Connes, Consani and Marcolli construct a descended trace pairing through a restriction-map cokernel and compare it with primitive cohomology [6, Sections 4.2 to 5]. Their realization includes the full zero divisor with multiplicities, and their positivity criterion separates the trace identity from the sign [6, Theorem 6.1 and Proposition 6.2]. This provides a closer arithmetic precedent than an abstract Hilbert-space normalization. The present extension obstruction concerns an extra orthogonality requirement; it is not a contradiction of that trace construction.

The semilocal Sonin isomorphisms in [7, Theorem 4.6] are topological isomorphisms, with distinct inherited Hilbert norms in the entire realization. Earlier work of Burnol connects Sonin spaces, co-Poisson summation and completed Mellin transforms [1, Introduction and Section 2]. The common entire carrier is therefore prior analytic structure. The elementary contribution here is the odd-order test for the particular positive multiplier proposed as a normalization of its metric transition.

A sheaf organization supplies another kind of compatibility. Connes's account places semilocal crossed products over $\text{Spec } \mathbb{Z}$ and describes their cohomological setting [3, Section 7.3, Theorem 7.2]. Restriction maps and a generic stalk specify how local arithmetic data fit together. A Hermitian pairing requires further duality and trace data, followed by a sign theorem. An identification of the resulting complex with the analytic Sonin carrier must respect those data and the trace comparison.

The adelic divisor construction of Connes and Consani uses rank-one torsion-free groups with possibly degenerate seminorms and identifies their classes as a Picard monoid [5, Definition 3.1 and Theorem 3.4]. The framed uniformization in that work provides geometric coordinates, not an arithmetic Hodge-index theorem for the Weil test form. In particular a monoid allowing singular

metrics cannot be treated as an already polarized abelian variety without an additional construction and proof.

Deninger's proposed arithmetic cohomology makes the desired sign and adjoint relation explicit through a Hodge-star pairing [8, Section 3]. A related weight identity is proved for a foliated three-manifold under stated geometric hypotheses [8, Theorem 4.1]. The arithmetic realization is a further conjectural step. This comparison explains why an independently constructed pairing is essential: once the positive pairing and the complete spectral realization are assumed, the centered spectral conclusion is a short operator argument.

These sources motivate a construction through arithmetic duality rather than prescribing the two unsuccessful repairs studied here. The local sign theorem excludes orthogonal signed increments under its stated hypotheses. It leaves open a geometric pairing whose primitive projection changes with support, whose intermediate forms are indefinite, or whose mixed terms reproduce local arithmetic contributions.

12.1 Chain maps and Fourier duality

Part I, Proposition 14.1, supplies a concrete algebraic repair of the nonmultiplicative shell tensor map. The two shell indicators e_0, e_1 individually induce algebra embeddings. Their signed combination extends to the Hochschild complex by taking the same linear combination of the two induced chain maps in every degree. This is different from tensoring the signed degree-zero map with itself. Its chain compatibility follows from multiplicativity of the two individual embeddings.

The corrected map also has a finite-prime chain square, as established in Part I, Section 15. These facts supply an algebraic correspondence before a primitive projection or pairing has been constructed. The maps act on uncompleted algebraic Hochschild chains. Applying a topological trace realization requires a specified completion and continuity statements for these maps.

The repair does not preserve the original componentwise Fourier symmetry in positive chain degree. Part I, Proposition 14.2, evaluates its degree-one coefficient on the small ball B_{-1}^2 . The coefficient itself vanishes there, whereas its local Fourier transform is $-(p-1)^3/p^2$. In degree $n \geq 1$, Part I, equation (92), gives the discrepancy

$$(p-1)^{n+1}(p^{-(n+1)} - p^{-1}) < 0$$

on B_{-1}^{n+1} . Both values are exact local-field calculations. The higher-degree failure cannot be repaired by citing the Fourier-fixed degree-zero shell alone.

This comparison distinguishes two uses of the word duality. On a Hilbert carrier, a bounded symmetry may commute with a compressed metric operator under the hypotheses in Section 4. On a Hochschild complex, the relevant duality must interact with its algebra product and boundary. Local Fourier transformation exchanges pointwise multiplication and additive convolution; it is not being assumed to be a chain map for either unmodified complex. A correspondence that is correct for one product therefore needs an explicit comparison with the dual product before its Fourier action can be used in a geometric pairing.

The chain construction is evidence that rejecting a naive tensor map need not reject an entire cohomological route. It also gives a concrete next test: identify a complex, duality and primitive projection for which the corrected chain correspondence has the required pairing defect. The local coefficient discrepancy specifies an equation that such a construction must account for.

13 Reformulated comparison

An arithmetic transition can be tested without discarding its correction terms. For a finite place set S and support radius a , suppose a specified complex has a primitive sector $P_{S,a}$, a Hermitian pairing $I_{S,a}$, and a complex-linear correspondence $\Gamma_{S,a} : \mathcal{T}_a \rightarrow P_{S,a}$. These are proposed geometric objects, not defined by pulling back the Weil form. Let a transition to (T, b) include an explicit map ι and a comparison of test images. For a common test f , write its image difference as $\eta f = \Gamma_{T,b}f - \iota\Gamma_{S,a}f$.

The exact pairing difference is then

$$\begin{aligned} & I_{T,b}(\Gamma_{T,b}f, \Gamma_{T,b}g) - I_{S,a}(\Gamma_{S,a}f, \Gamma_{S,a}g) \\ &= E(f, g) + I_{T,b}(\iota\Gamma_{S,a}f, \eta g) + I_{T,b}(\eta f, \iota\Gamma_{S,a}g) + I_{T,b}(\eta f, \eta g), \end{aligned} \quad (46)$$

where E is the old-image defect defined by (41) with support indices included. An arithmetic theorem must identify the right side with the relevant prime sum and any boundary or degree term. Deleting the mixed terms is justified only by a proved orthogonality identity.

For a one-prime intermediate transition, an admissible comparison target has the form

$$\Delta I(f, g) = W_p(f * g^*) + C_{S,p,a}(f, g), \quad (47)$$

where $C_{S,p,a}$ is an explicitly constructed correction. The equation is useful only if $C_{S,p,a}$ is determined by the geometry and its behavior under further transitions is proved. Defining C as the unexplained difference of the two sides would merely restate the problem.

At a support-complete stage the desired sign is $I_{S,a}(x, x) \leq 0$ on the appropriate primitive sector. The full comparison must include the archimedean distribution and the pole contribution, or reconstruct them from an explicit degree complement. If it gives

$$B_W(f, g) = -I(\Gamma f, \Gamma g) \quad (f, g \in \mathcal{T}) \quad (48)$$

with nonpositive I , then $B_W(f, f) \geq 0$ on every complex compact smooth test. The classical Weil criterion then implies RH. Equation (48) and its geometric sign are the required outcome of the program. Neither is a consequence of equivalent Sonin norms.

The obstructions proved here impose concrete restrictions on attempts at this outcome. Positive pointwise polar normalization cannot preserve the entire carrier. Intrinsic normalization must be analyzed through its compressed operator and its test-map equations. An orthogonal nonpositive local enlargement cannot realize the exact intermediate $+W_p$ increment on the full pole-moment kernel. Support-complete sign theorems and correspondences with computed mixed terms remain compatible with these results.

14 Formal arithmetic statements

The accompanying Lean development includes the arithmetic coefficient calculations and the squared entire identity obstruction. The companion sources are at <https://github.com/YonedaAI/adelic-polarization-conjecture>. The formalization uses Lean v4.30.0-rc2 and the pinned mathlib revision 0f9072dd907c6e2e4264ab241a049cab50137f7c. Its theorems rely only on the usual mathlib logical foundations, with no project arithmetic axioms or admitted proofs. The analytic applications below remain separate from the finite algebra they use.

The module `lean/APC/LocalSigns.lean` defines the finite sum in (22) for real coefficients and the coefficient polynomial evaluation as an actual finite sum. The theorems `APC.minus_exact` and `APC.plus_exact` establish the two values at 11. Theorems with suffixes `minus_negative` and

`plus_positive` establish their strict signs. The four radius and inverse-radius moment theorems prove the zero polynomial evaluations for the two vectors. The moment identities are conclusions of these proofs, not assumptions in a sign certificate.

The module `lean/APC/LatticeDistribution.lean` extends the finite calculation to a translated profile

$$H_c(x) = \sum_{i,j} c_i c_j H(x - (i - j) \log p). \quad (49)$$

Under the explicit premise that $H(k \log p)$ equals A at $k = 0$ and zero at every other integer k , its sampling theorem identifies the prime-lattice values of H_c with A times the discrete coefficient correlation. A further theorem derives this premise from $H(0) = A$ and $H(x) = 0$ for $|x| \geq \log p$. The module proves stability of the finite prime-power sum once the cutoff exceeds the coefficient-vector length, and its theorems `APC.translated_minus_exact` and `APC.translated_plus_exact` give the exact local distribution values for the two profiles at 11.

In Proposition 6.1, $H = \psi * \psi^*$ and $A = \|\psi\|_2^2 > 0$. The Lean profile premise does not itself construct this smooth autocorrelation or prove its norm positive. The compact smooth bump, convolution integral and Laplace moment identities are proved in the paper. The all-prime packet formula and the constrained four-bump signature are also paper proofs. Thus the formal finite values support, but do not replace, the complete smooth sign theorem.

The module `lean/APC/EntireObstruction.lean` defines the actual entire function Δ_p and proves its analyticity, its zero at zero and the derivative $\log p(1 - p^{-1})$ for every natural number $p > 1$. The theorem `APC.eulerDenominator_order_one` deduces its exact analytic order. The final theorem `APC.eulerDenominator_no_entire_square` rules out $\Delta_p G^2 = F^2$ for nonzero functions analytic at every complex point. Finite order is deduced from the analytic identity theorem and is not retained as an additional premise of the entire theorem.

The passage from almost-everywhere boundary equality to the entire squared identity is supplied by Lemma 3.3. The formal module does not define the weighted Hilbert carrier, its orthogonal projection or its polar decomposition. Those operator constructions and the geometric extension obstruction are established by the proofs in this paper. No formal theorem asserts an arithmetic polarization or the sign of the full Weil form.

15 Conclusion

The specified ambient normalization fails by a local holomorphic parity argument, and the specified orthogonal extension fails on exact arithmetic smooth witnesses. The compressed Sonin normalization and the support-indexed geometric program survive these tests with additional comparison obligations. A successful construction must derive its pairing, retain mixed and degree terms, and prove the full arithmetic identity together with its sign.

References

- [1] Jean-François Burnol. *Two complete and minimal systems associated with the zeros of the Riemann zeta function*. arXiv:math/0203120v7, 2004. <https://arxiv.org/abs/math/0203120v7>.
- [2] Alain Connes. *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*. arXiv:math/9811068v1, 1998. <https://arxiv.org/abs/math/9811068v1>.
- [3] Alain Connes. *The Riemann Hypothesis: Past, Present and a Letter Through Time*. arXiv:2602.04022v1, 2026. <https://arxiv.org/abs/2602.04022v1>.

- [4] Alain Connes and Caterina Consani. *Weil positivity and Trace formula, the archimedean place*. arXiv:2006.13771v1, 2020. <https://arxiv.org/abs/2006.13771v1>.
- [5] Alain Connes and Caterina Consani. *On the Jacobian of $\overline{\text{Spec } \mathbb{Z}}$* . arXiv:2602.15941v1, 2026. <https://arxiv.org/abs/2602.15941v1>.
- [6] Alain Connes, Caterina Consani and Matilde Marcolli. *The Weil proof and the geometry of the adeles class space*. arXiv:math/0703392v1, 2007. <https://arxiv.org/abs/math/0703392v1>.
- [7] Alain Connes, Caterina Consani and Henri Moscovici. *Zeta zeros and prolate wave operators: Semilocal adelic operators*. arXiv:2310.18423v2, 2024. <https://arxiv.org/abs/2310.18423v2>.
- [8] Christopher Deninger. *Arithmetic Geometry and Analysis on Foliated Spaces*. arXiv:math/0505354v1, 2005. <https://arxiv.org/abs/math/0505354v1>.
- [9] Matthew Long, The YonedaAI Collaboration. *Prime-Adjoining Correspondences*. Part I of *Adelic Polarization Conjecture*, 2026. Companion manuscript, `local-correspondence.tex`.